

PYTHON FULL STACK TRAINING: TUPLE LOGIC & INTERVIEW PREP

Topic: Tuples, Immutability, and Data Integrity

--- PART 1: BASICS & INITIALIZATION ---

1. Create a tuple containing five different data types (int, float, string, list, boolean).
2. Write a script to check the type of a tuple with a single element. Show the difference between (5) and (5,).
3. Access the last element of a tuple without knowing its length.
4. Access the second to last element of a tuple using negative indexing.
5. Given `nested_tuple = ("Python", [10, 20, 30], (5, 15, 25))`, print the number 20.
6. Check if the element 'Sreeraj' exists in a tuple using a membership operator.
7. Find the memory size of a list vs. a tuple with the same elements using the sys module.
8. Unpack a tuple of 3 elements into three variables: x, y, and z.
9. Demonstrate what happens if you try to unpack a tuple of 4 elements into 3 variables.
10. Use the 'extended iterable unpacking' (using *) to grab the first element and the rest into a list.

--- PART 2: IMMUTABILITY LOGIC ---

11. Write code that attempts to change the first element of a tuple and handle the resulting TypeError gracefully.
12. Given a tuple: `t = (1, 2, [3, 4])`. Change the value 3 to 30. Explain why this works despite tuples being immutable.
13. Create two tuples, concatenate them, and assign them to a new variable.
14. Use the repetition operator (*) to create a tuple of ten zeros.
15. Swap two variables a and b using tuple unpacking logic in a single line.
16. Write a program to "add" an item to a tuple by converting it to a list first.
17. Write a program to "remove" an item from a tuple.
18. Delete an entire tuple variable from memory and verify its absence using a try-except block.
19. Sort a tuple of integers and return the result as a new tuple.

20. Reverse a tuple using the slicing method [::-1].

--- PART 3: METHODS & AGGREGATION ---

21. Find the index of the first occurrence of the number 10 in a tuple.

22. Count how many times the string "Python" appears in a tuple of job roles.

23. Find the maximum and minimum values in a tuple of stock prices.

24. Calculate the sum of all numeric elements in a tuple.

25. Given a tuple of tuples: ((1, 2), (3, 4), (5, 6)), calculate the sum of the second element of each internal tuple.

26. Use the index() method to find the position of 'Apple' starting from index 3 in a large fruit tuple.

27. Write a function that takes a tuple and returns a new tuple containing only the even numbers.

28. Convert a list of tuples into a single flat list.

29. Create a tuple from a user-input string where each character is an element.

30. Check if all elements in a tuple are truthy using the all() function.

--- PART 4: SLICING & LOOPING ---

31. Extract a sub-tuple containing the elements from index 2 to 5 (inclusive).

32. Use slicing to get every second element of a tuple.

33. Write a 'for' loop to print each element of a tuple with its corresponding index number.

34. Use a 'while' loop to iterate through a tuple backwards.

35. Create a tuple of 10 numbers and slice it to get the last 3 elements.

36. Slice a tuple to remove the first and last elements.

37. Use a for loop to concatenate all strings in a tuple into a single sentence.

38. Compare two tuples (1, 2, 3) and (1, 2, 4). Explain the logic of how Python compares them.

39. Write a program to find duplicate elements in a tuple.

40. Zip two tuples together to create a list of coordinate pairs.

PART 5 – Logical Tuple Problems

31. Create a tuple of numbers and **find the second largest number**.
32. Write a program to **remove duplicate elements from a tuple**.
33. Write a program to **find the frequency of each element in a tuple**.
34. Write a program to **convert a list of tuples into a dictionary**.

Example:

[(1,'A'), (2,'B'), (3,'C')]

35. Write a program to **sort a tuple of numbers**.
36. Write a program to **sort a tuple of tuples based on the second element**.

Example:

((1,5),(3,2),(4,8))

37. Write a program to **find common elements between two tuples**.
38. Write a program to **convert a tuple into a string**.
39. Write a program to **count vowels present in a tuple of characters**.
40. Write a program to **find the product of all numbers in a tuple**.

--- PART 6: INDUSTRIAL TRENDS & ADVANCED (MISSING TOPICS) ---

41. [NamedTuple] Use 'collections.namedtuple' to create a 'Student' object with fields 'name', 'roll_no', and 'course'.
42. [Dictionary Keys] Demonstrate using a tuple as a key in a dictionary (e.g., latitude/longitude). Explain why a list cannot be a key.
43. [Data Integrity] Write a function that returns multiple values (name, age, salary) as a tuple.
44. [Efficiency] Iterate through a tuple of 1,000,000 integers and measure the time taken. Compare it with a list.
45. [Nested Logic] Given a list of tuples representing products: [("Laptop", 1000), ("Mouse", 25)]. Sort the list based on the price (the second element).
46. [String Formatting] Use a tuple to provide values for an old-style string formatting: "Hello %s, your ID is %d".
47. [Function Arguments] Write a function that accepts *args and prints the type of 'args'.

48. [Hashing] Use the `hash()` function on a tuple. Then try it on a list. Note the difference.
49. [Conversion] Convert a tuple of single-character strings into a single string.
50. [Filtering] Use the `filter()` function with a lambda to remove all `None` values from a tuple.